

Introduction

Over the Air Computing offers a paradigm shift in how distributed computations are performed in wireless networks. By exploiting the inherent properties of the wireless medium, Over-the-Air Computation (OAC) enables simultaneous transmissions from multiple devices to be combined in a way that directly computes a desired function of the transmitted data. This technique has garnered significant attention due to its potential to enhance efficiency and reduce latency in applications such as federated learning, distributed sensing, and real-time analytics. The motivation behind this thesis is to explore the practical implementation of OAC using Software Defined Radios Software Defined Radio (SDR) and MATLAB, addressing the challenges associated with synchronization, channel impairments, and waveform design. The research objectives include designing a prototype OAC network, validating its performance, and analyzing the impact of various impairments on computation accuracy.

0.a Background and Motivation

The increasing need for efficiency in computation-oriented applications where the interest lies in a function of local information, rather than the information itself [1]. The conventional approach of separating communication and computation often proves suboptimal [1], [2], [3]. The potential of OAC Computation is to achieve significantly higher computation rates by harnessing interference through signal superposition in a wireless Multiple Access Channel (MAC) [1].

0.b Problem Statement and Research Objectives

Focusing on designing and validating a practical OAC network layout suitable for implementation in a controllable environment[1]. The specific role of Software Defined Radios (SDRs) and MATLAB/simulation tools for prototype validation and addressing practical limitations such as synchronization and channel impairments [1], [4], [5].

0.c Thesis Contributions and Organization

References

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